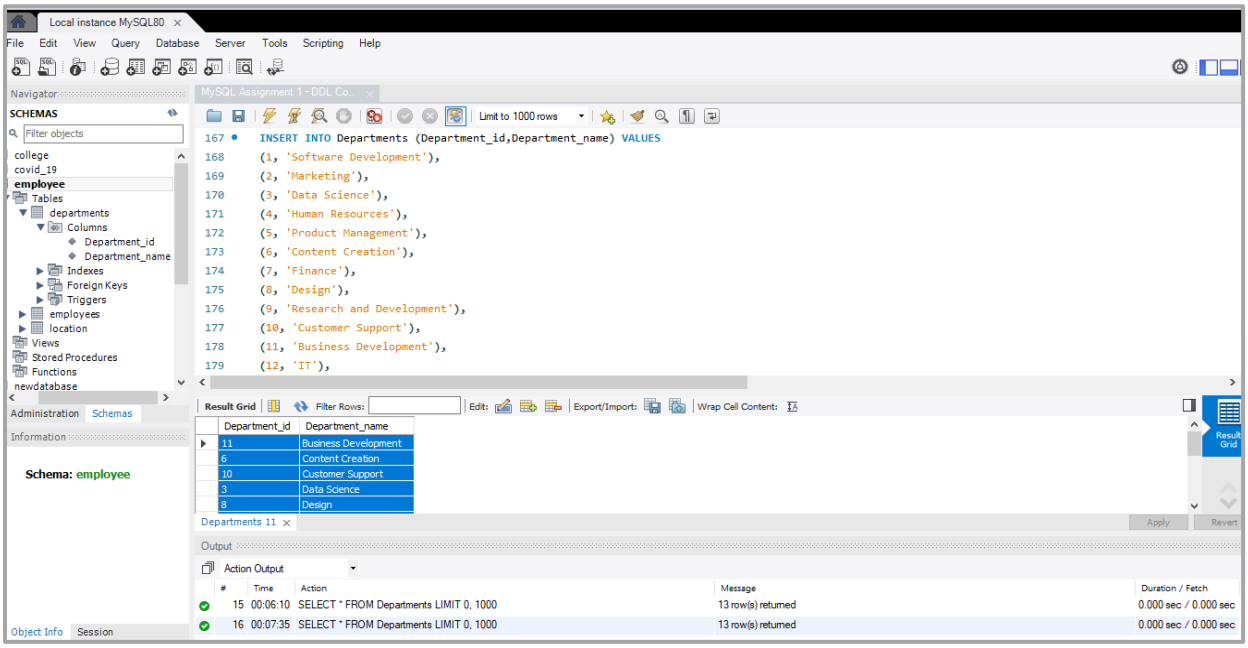
MySQL Assignment 2 – Querying Data

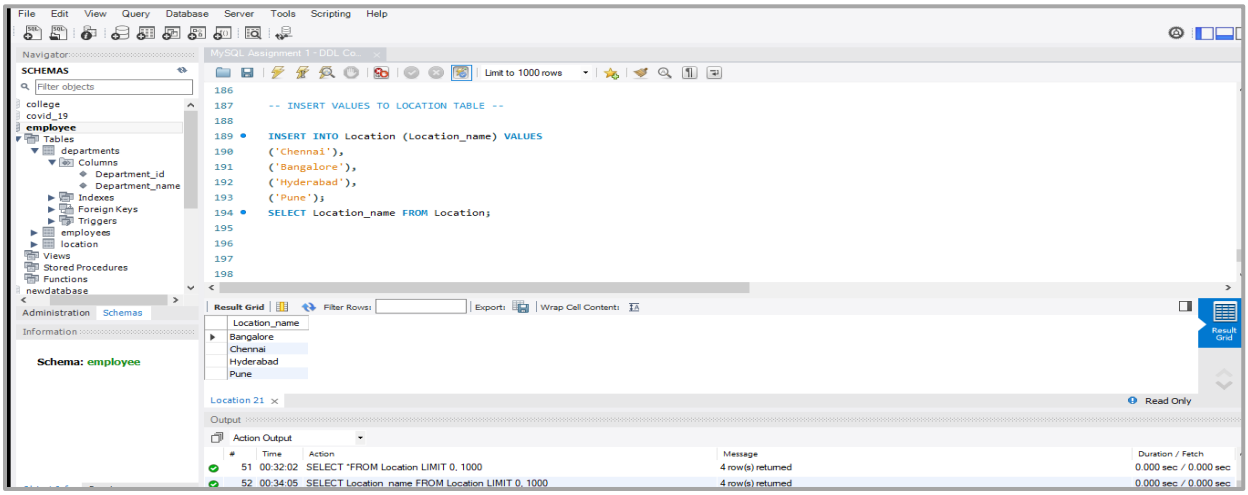
Submitted by: RESHMA V M

★ Use the provided DML insert codes to populate the tables from here: Employee Data

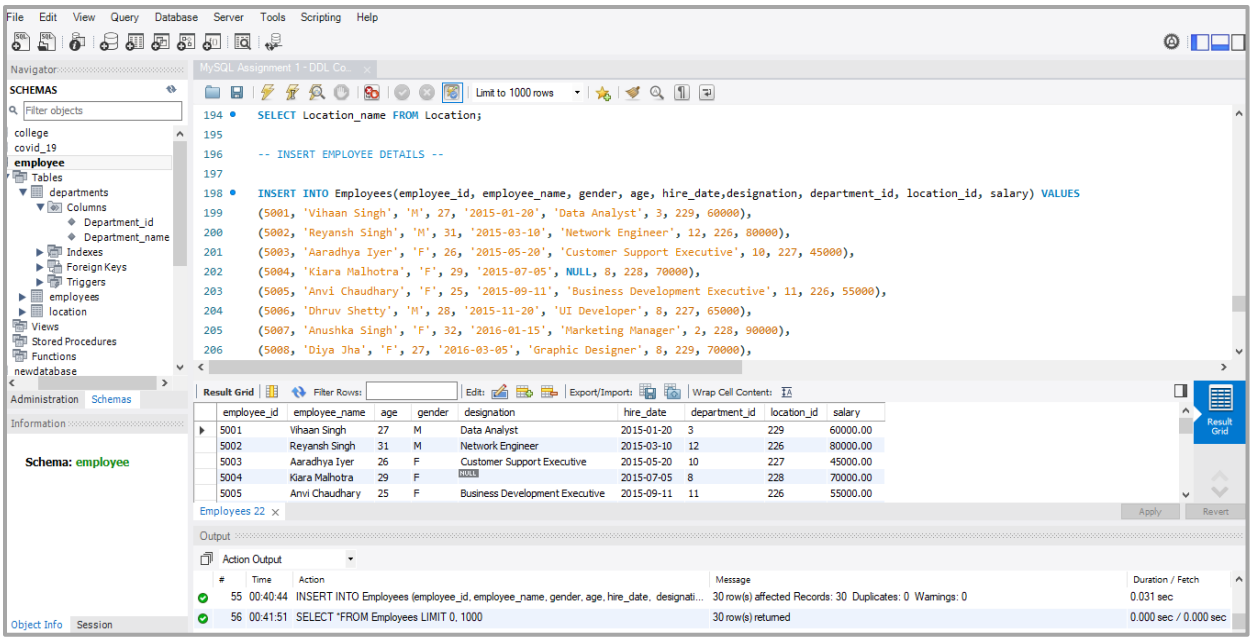
★ Execute the queries in this assignment after inserting the data.



Pic: Departments table



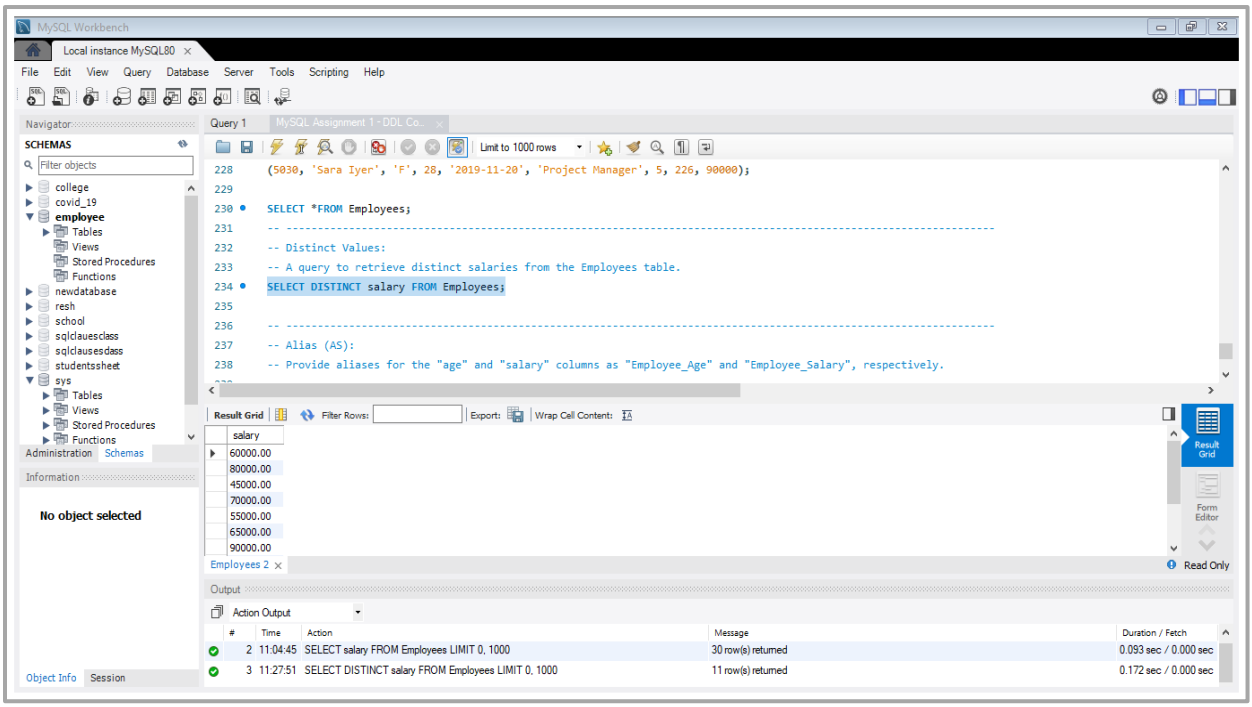
Pic: Location table



Pic: Employees table

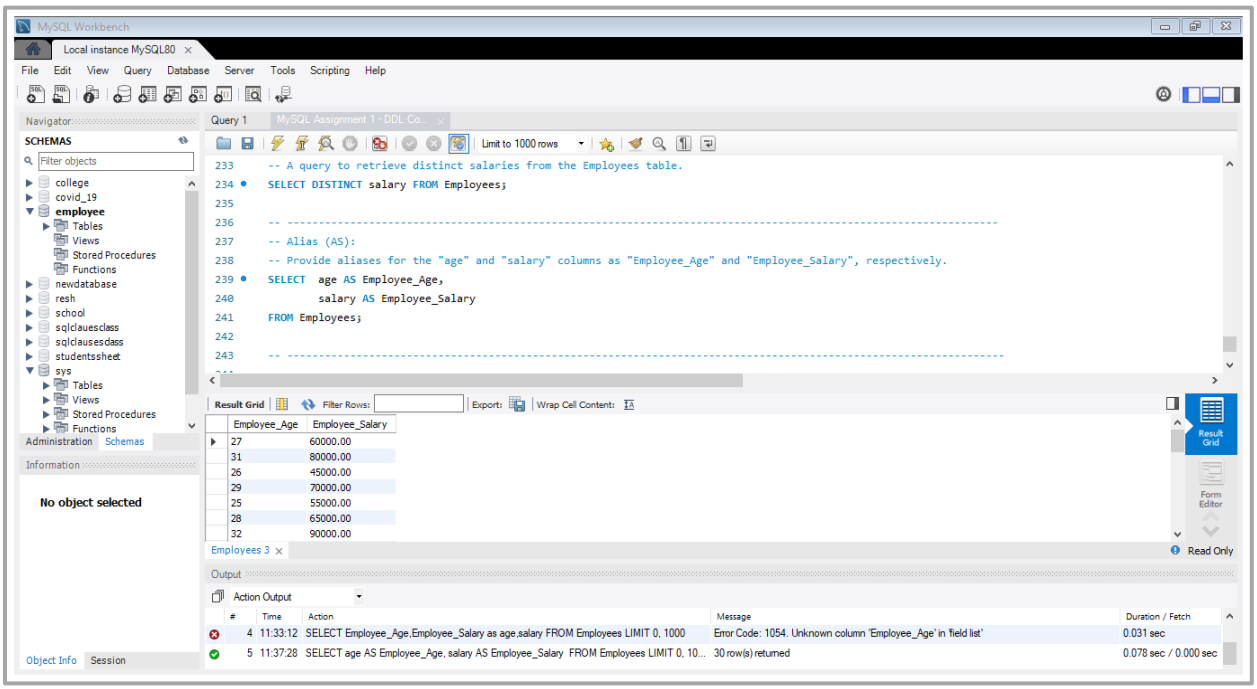
1. Distinct Values

- A query to retrieve distinct salaries from the Employees table.



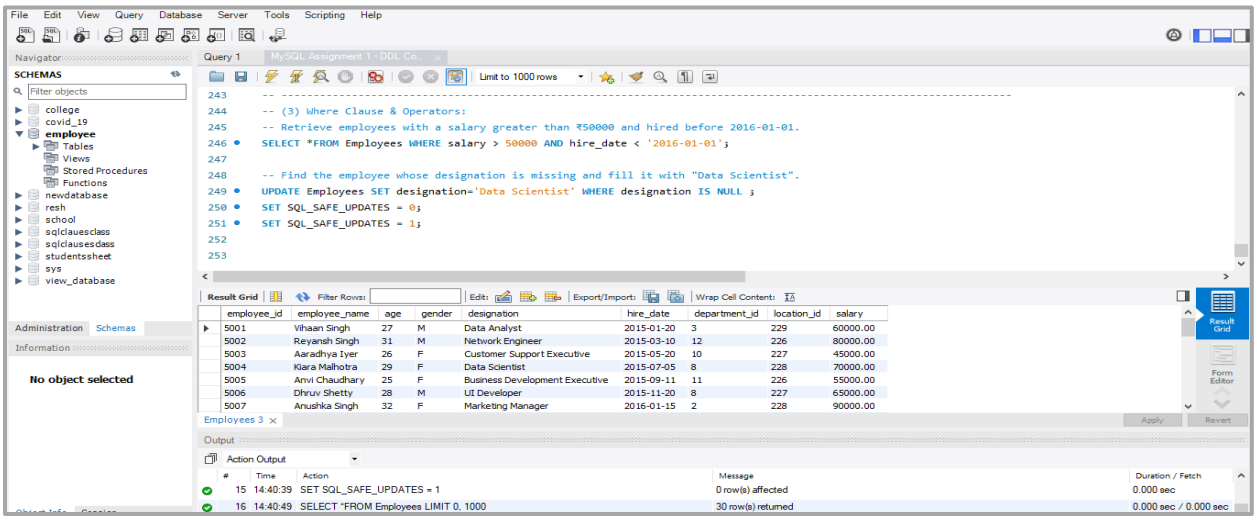
2. Alias (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.



3. Where Clause & Operators

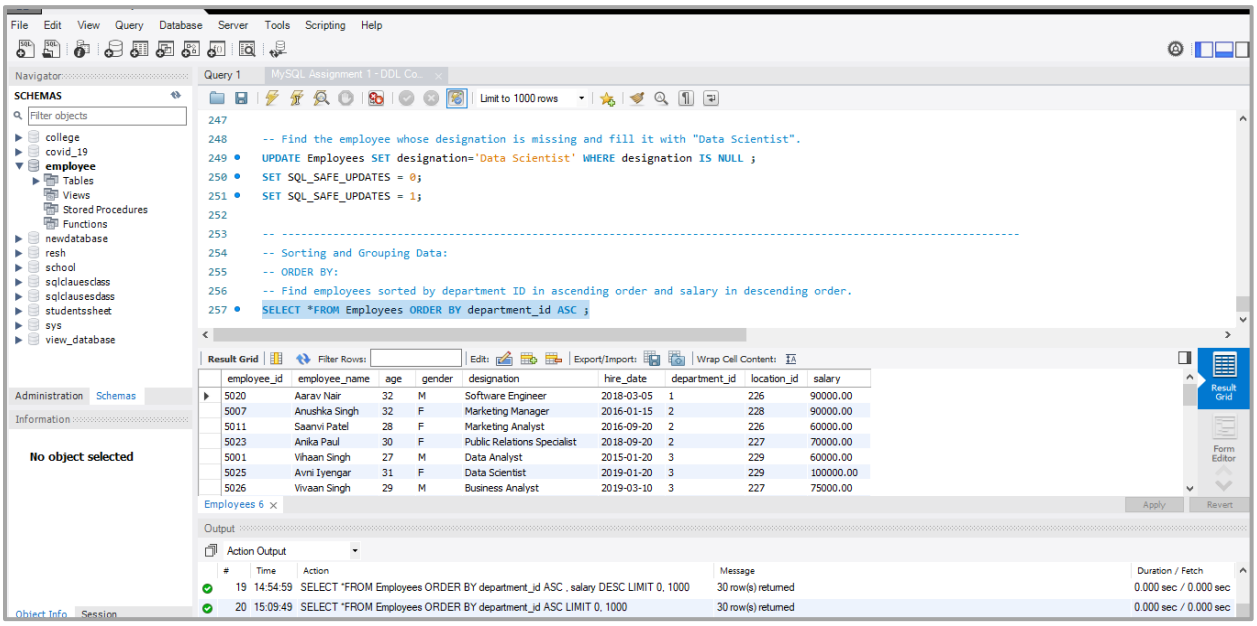
- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
- Find the employee whose designation is missing and fill it with "Data Scientist".



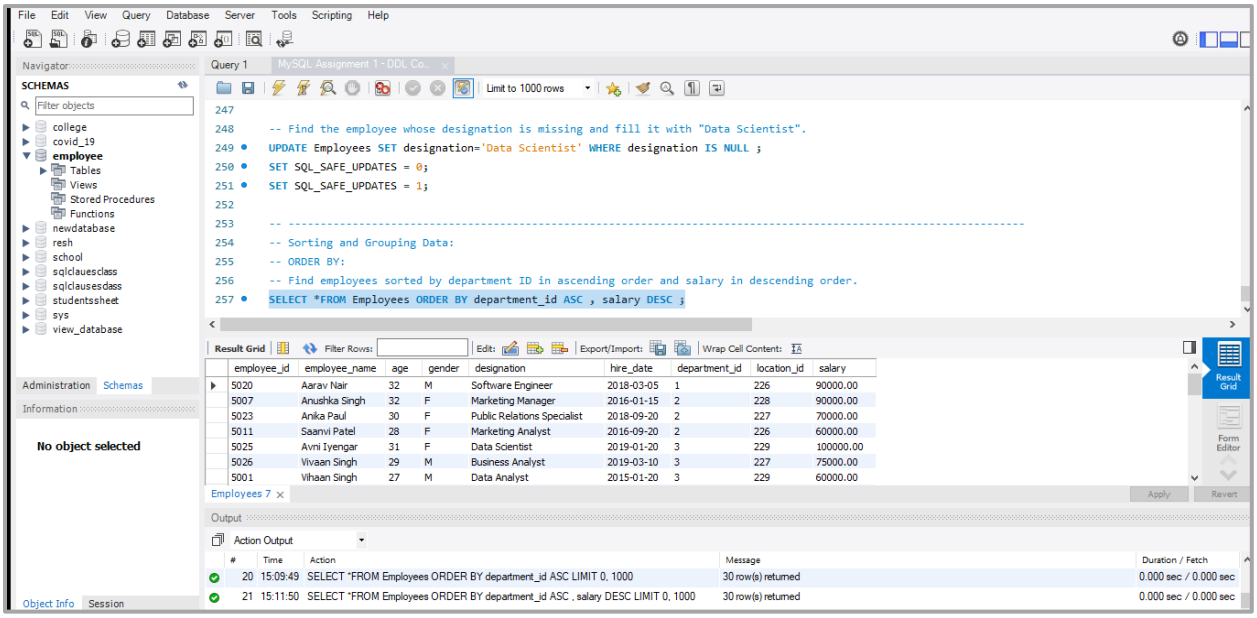
Sorting and Grouping Data:

1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.



Pic : Employees sorted by department ID in ascending order



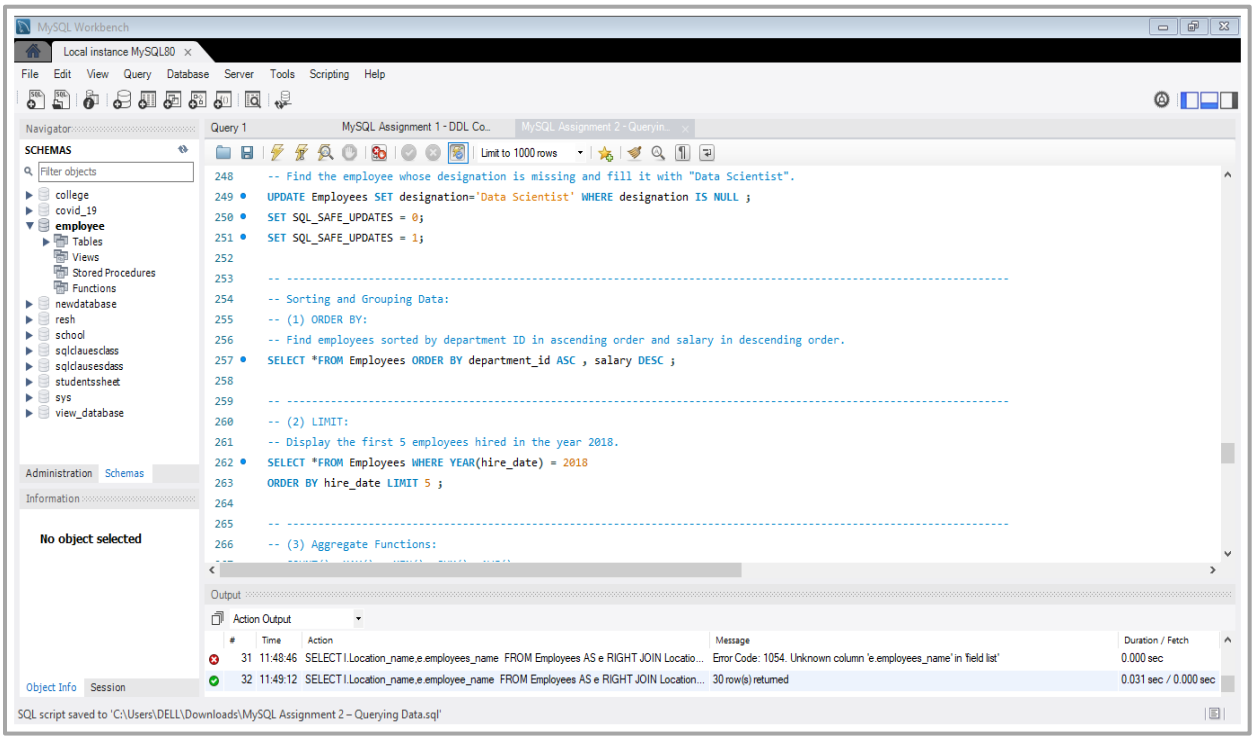
Pic : Employees sorted by department ID in ascending order & salary in descending order

salary DESC sorts employees by salary only within the same department ID

1. LIMIT:

LIMIT in MySQL is used to restrict the number of rows returned by a query.

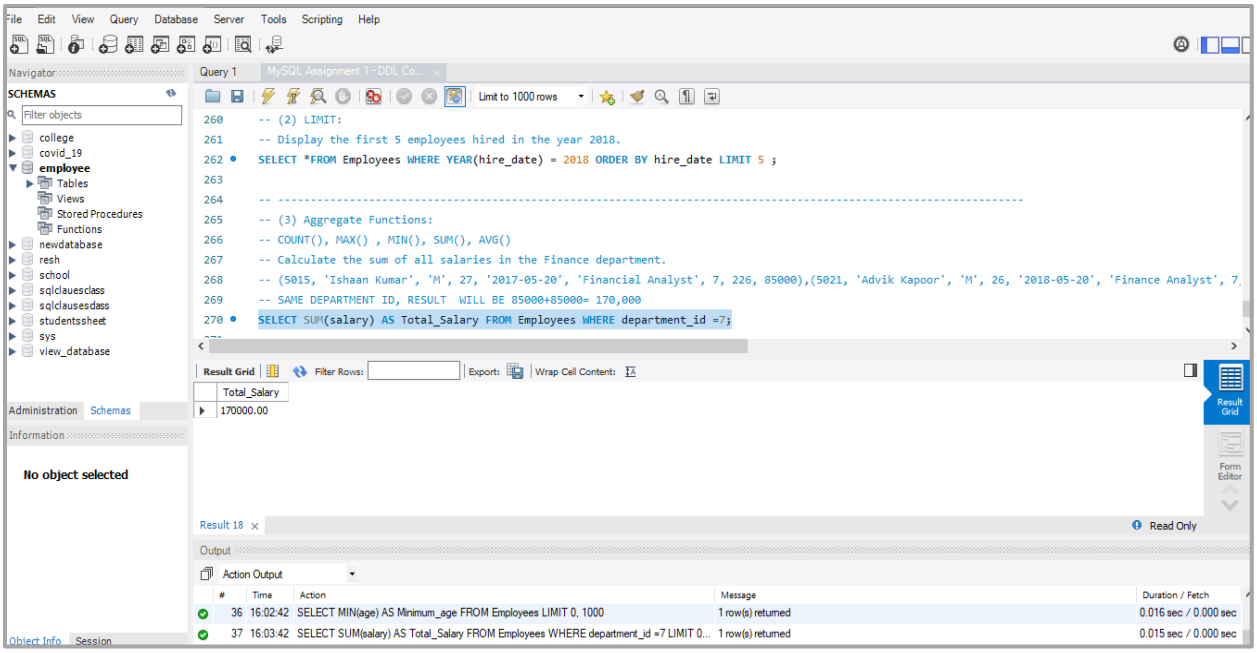
- Display the first 5 employees hired in the year 2018.



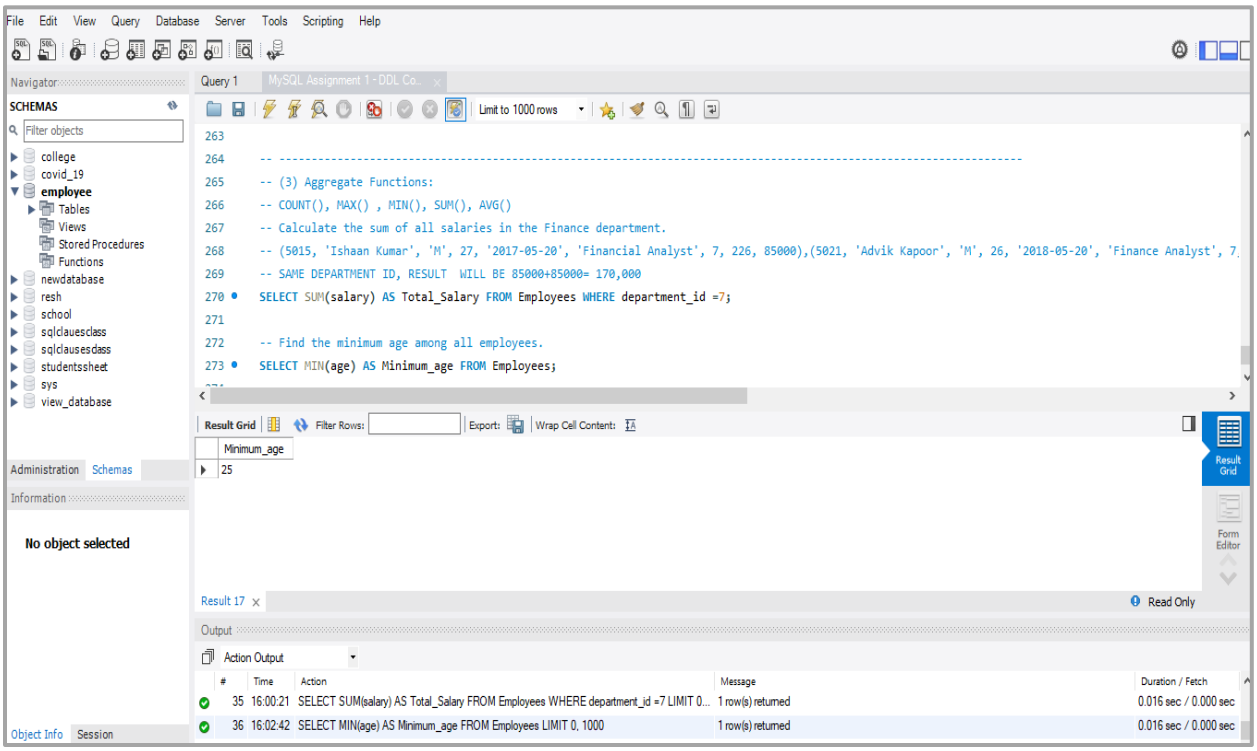
2. Aggregate Functions:

- ❖ Aggregate functions in MySQL are used to perform calculations on multiple rows and return a single result.
- ❖ The main aggregate functions are COUNT(), SUM(), AVG(), MAX(), and MIN().
- ❖ They are commonly used with GROUP BY and HAVING clauses.

- Calculate the sum of all salaries in the Finance department.

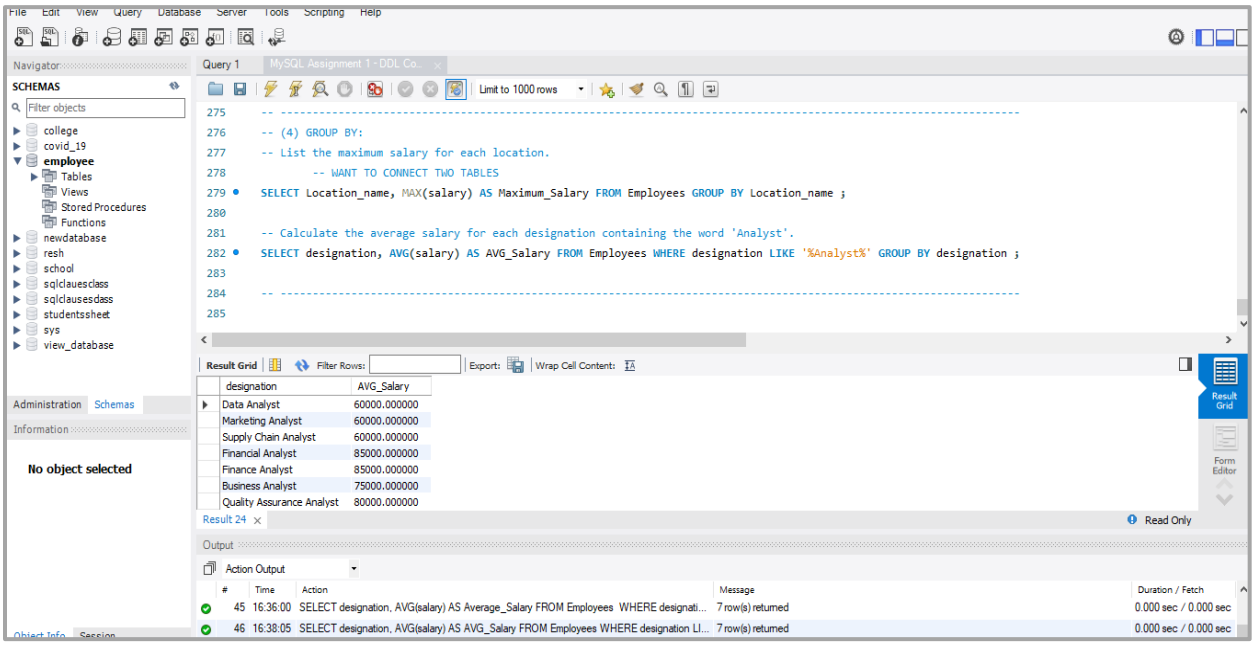


- Find the minimum age among all employees.



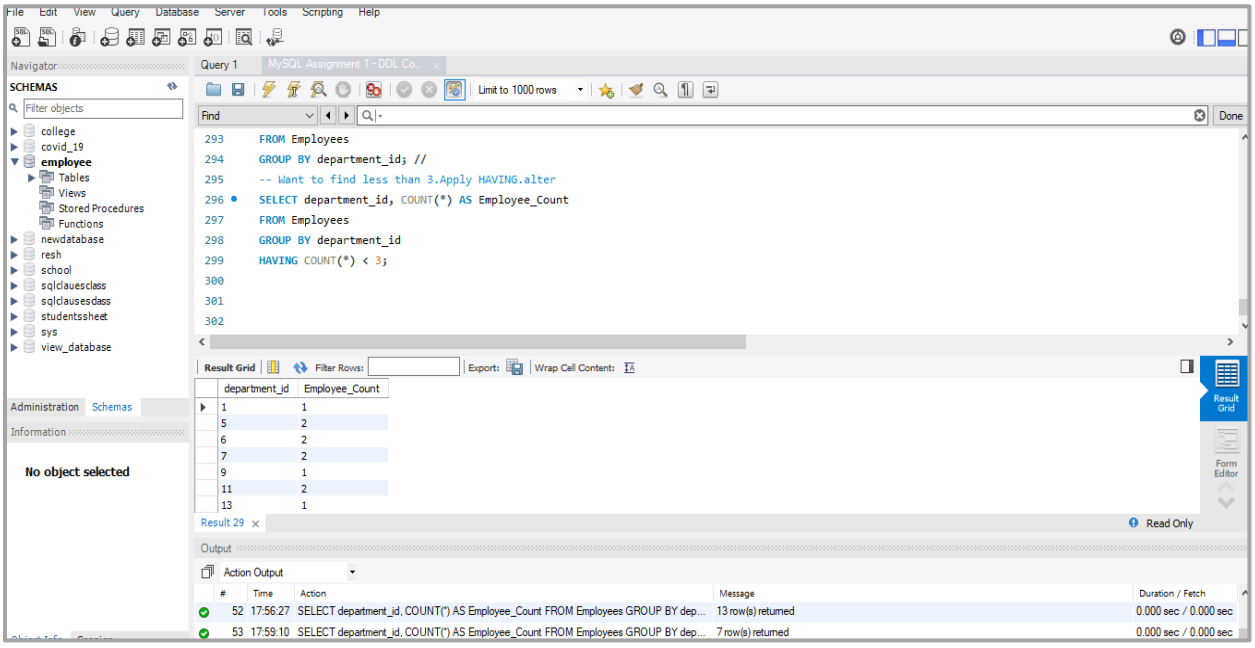
GROUP BY:

- List the maximum salary for each location.
- Calculate the average salary for each designation containing the word 'Analyst'.

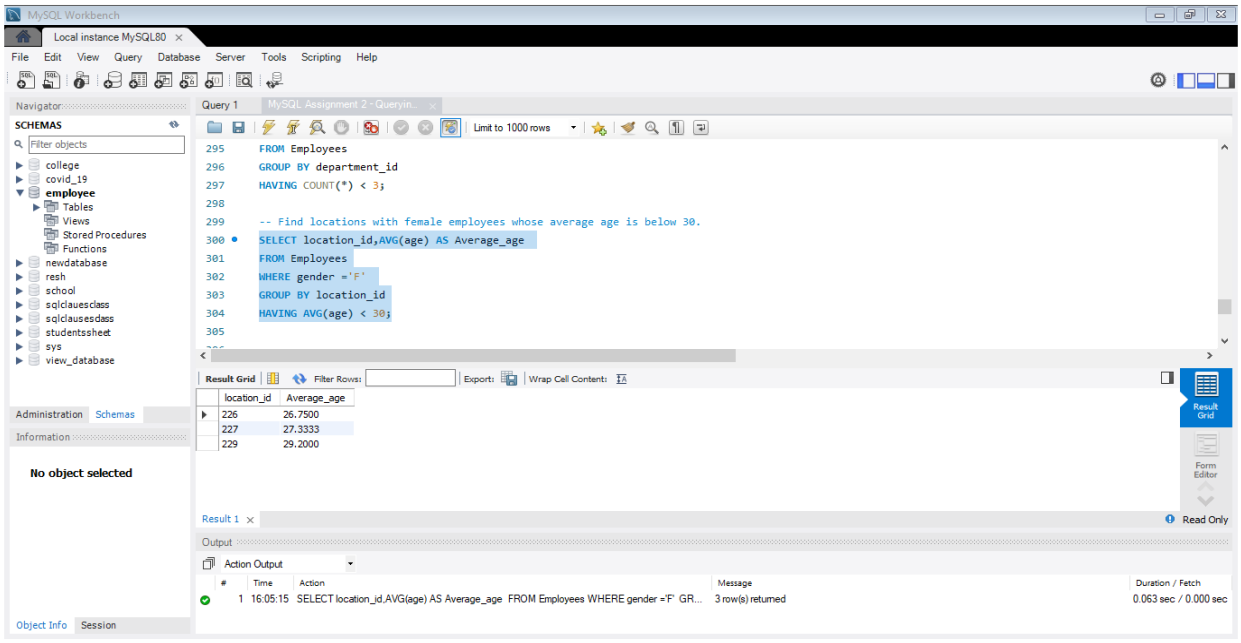


HAVING:

- Find departments with less than 3 employees.



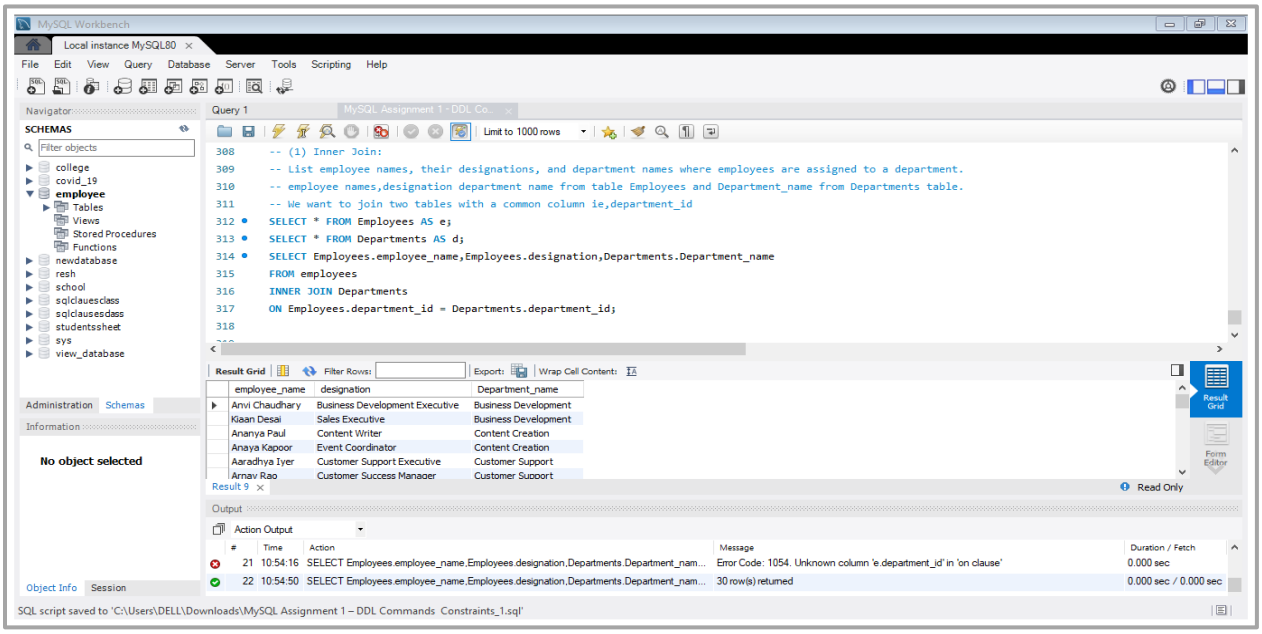
- Find locations with female employees whose average age is below 30.



Joins:

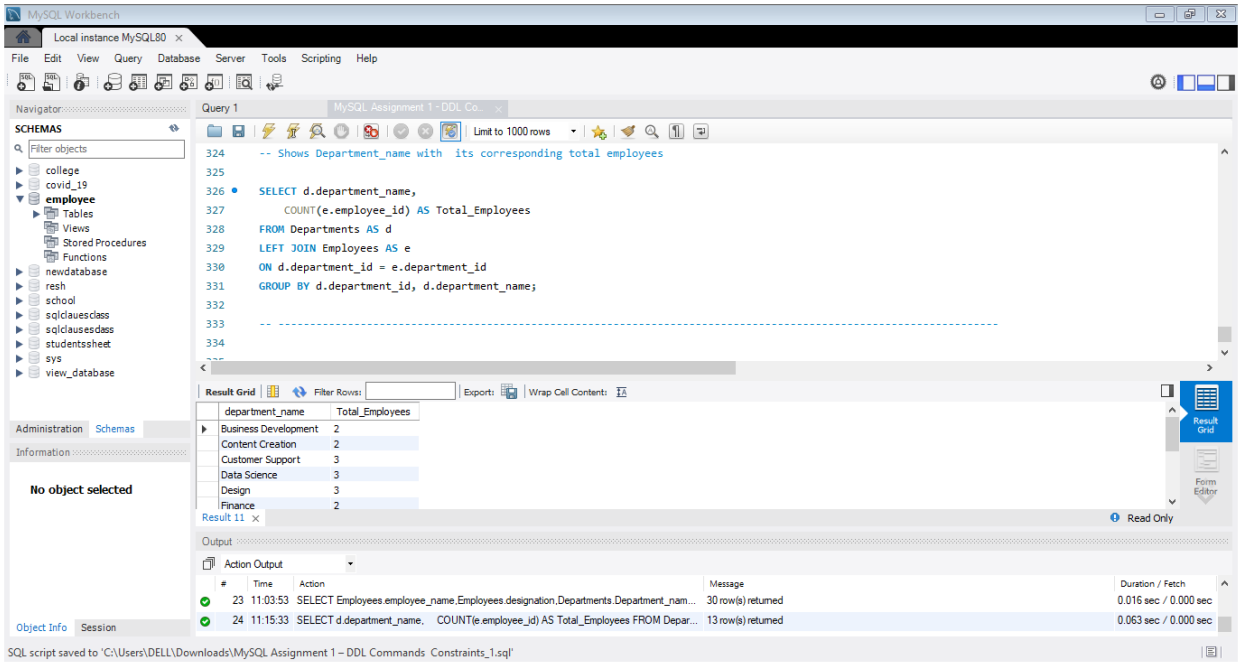
Inner Join:

- List employee names, their designations, and department names where employees are assigned to a department.



Left Join:

- List all departments along with the total number of employees in each department, including departments with no employees.



Right Join:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

